Maths for AI

(Calculus) –

- f(x → y à learning (?) : mathematicl that takes continuous numerical input

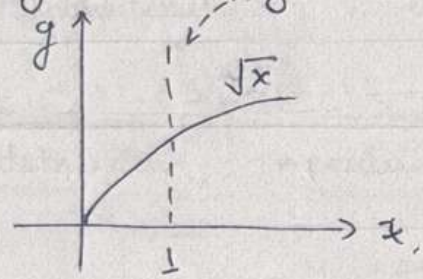
F(x : Each input has one & only one 1/p (cdomain Range)

Domain: All valid input values beneath function rule

Range: All possible 1/p values f(x) for thoe domain numbers

e,g $f(x) = x^2$

grad(?)



* Nested functions
or composite function

e.g $f(g(x))$ rc. $f(x)$

$g(x) = x+5$ $\sqrt{x+5}$

Domain: $[5, \infty)$

soln' ⇒ $x+5 \geq 0$

Domain: $[5, \infty]$

* $f(x) = f(x) \Longrightarrow \sqrt{x+5}$

$f(g(x)) = [5, \infty)$

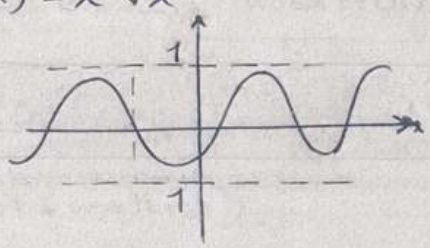
Domain: $[5, \infty)$

how fx behaves

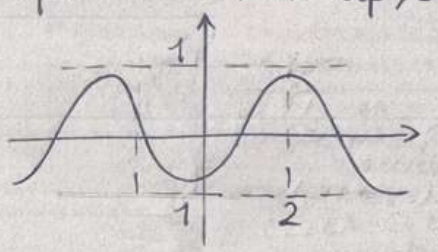
- Scaler multiplies centre width bro.
- Scaler delation lifts amplitude bx (Shats upion evel)

e.g $f(x) = x^2$ = 3rd)

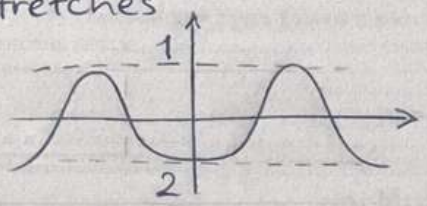
$g(x) = x\sqrt{x}$



*) Scaler dilation lifts amplitude stats up/down



*3) Scaler elevates it even stretches



*) Scaler elevates it even stretches.

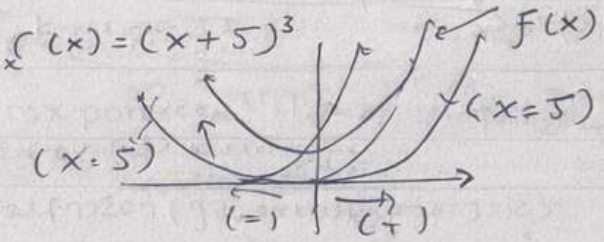
- Scaler adeletion with fnx

  re $f(x) = e$

$$\begin{cases} C'(x) = e^{-x} \ (k > 0) \\ C'(x) = C(x) \text{ more down} \end{cases}$$

eg $b(x) = \sin(kx) + 1$

$b(x) = 2\cos(x - \frac{\pi}{2})$



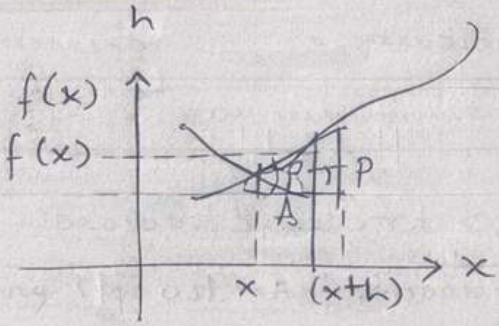
Differentiation

(sbscibeibes how fast one quantity changes compared to another

Ex: how fast y changes when x changes by $\frac{dy}{dx}$

$$y = f(x) \Rightarrow \frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

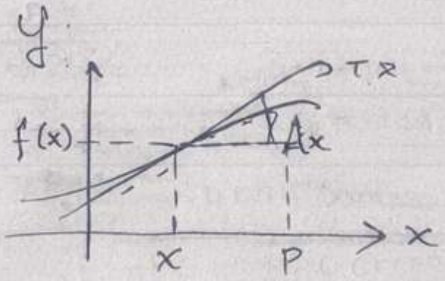
$$\frac{dy}{dx} = \frac{\text{change in } y}{\text{change in } x}$$



- $\frac{d}{dx} x^2 \longrightarrow c^2 x$

- $y = \sin(x) \rightarrow dy = \cos(x)$

Differentiation Rule



- Constant Rule

  $C \Rightarrow C' = 0$

- Product Rule (uv)

$$\frac{d}{dx}(uv) = u'v + u\frac{dv}{dx}$$

$$\left[\frac{d}{dx}(uv) = u'v + u\frac{dv}{dx}\right]$$

$$\frac{d}{dx}\left[\frac{f(x)f(x)-f(x)g'(x)}{(g(x)^2}\right.$$

eg $f(x)=\frac{\sin x}{\cos x}$, $g(x)=\cos x$

$$\frac{d}{dx}\left[\frac{\sin x}{\cos^2 x}\right]=\frac{\cos x\cdot\cos x-(-\sin x\sin x)}{\cos^2 x}$$

$$=\frac{\cos^2 x+\sin^2 x}{\cos^2 x}$$

$$\equiv \sec^2 x$$

Chain Rule

$$\frac{dy}{dx}=\frac{dy}{du}\cdot\frac{du}{dx}$$

a) $y=(2x+3)^2$

Let $u=2x+3$

$$\frac{du}{dx}=2$$

$$\frac{dy}{dx}=2(2x+3)\cdot 2$$

$$=8x+12$$

$$\frac{dy}{dx}=2u\cdot\frac{du}{dx}$$

$$\frac{du}{dx}=2$$

$$\frac{dy}{dx}=3(2x+3)^2\cdot 2$$

$$=6(2x+3)^2$$

$$=6(x^{2^2}+3x+3^2)$$

$$=12x^2+33x+27$$

b) $y=(2x+3)^3$

Let $u=2x+3$

$$\frac{du}{dx}=3u^2\frac{du}{dx}$$

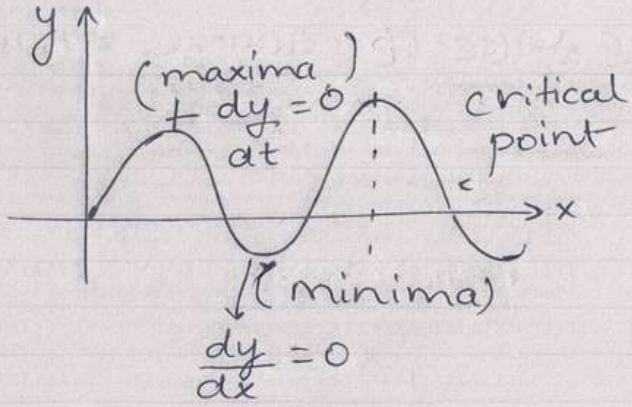
$$\frac{dy}{dx}=3(2x+3)^2\cdot 2$$

$$=6(cx^2+3x+3^2)$$

$$=12x^2+33x+27$$

Finding minima/maxima

- minima - the lowest value of a fnx
- maxima - the highest value of a fnx



- Ex: $f(x) = x^2 - 4 + 5 + g$

$$\frac{dy}{dx} = 2x - 4$$

$$\boxed{x = 2}$$

$$\boxed{\frac{d^2y}{dx^2} = 2}$$

i) $x = 2,\ y = ?$

ii) $x = 2,\ y = 1$

Note

i) +ve → minima

ii) −ve → maxima

$$\frac{7}{dx} = \frac{y}{2}$$

$$\frac{dy}{dx} = \frac{1}{dx} = 2x + 3$$

$$= x = 2$$

Note

if +ve ⟹ minima

ii) −ve ⟹ maxima

↓ negative value

(Global concave fnd)

(calculate using $\frac{d^2y}{dx}$ fnx)

O practice problem

O Find critical point for $f(x) = x^3 - 6x^2 + 9x + 1$